$0, 1, \ldots, \nu$. (This is a simplifying assumption the removal of which would require somewhat more intricate assumptions regarding the differential equations.)

With these assumptions in hand, begin from (2.9) and write

$$z_{n+1} - z(t_{n+1}) = \mathscr{G}_h(z_n) - \mathscr{G}_h(z(t_n)) + \mathscr{G}_h(z(t_n)) - \mathscr{F}_h(z(t_n)),$$

then take norms and use the triangle inequality and (2.10), (2.11) to get the following recurrent inequality for the error $\varepsilon_n = \|z_n - z(t_n)\|$:

$$\varepsilon_{n+1} \leq (1 + Lh)\varepsilon_n + \bar{K}h^{p+1}.$$

From this, the bound

$$\varepsilon_n \leq \frac{\bar{K}}{L} e^{Lnh} h^p, \qquad n = 0, 1, \ldots \nu, \tag{2.12}$$

follows by a straightforward calculation, for $h \leq \Delta$.

The assumption (2.10) that the local error is of order $p + 1$, $p > 0$, is termed the *consistency* of the numerical method. We say the method is *consistent of order p*.

The assumption (2.11) that the method does not increase the separation between two nearby trajectories by more than a factor of the form $1 + hL$ in each step is referred to as the *stability* of the method.

This result shows that a method which is consistent of order p and stable is convergent of order p.

Example 2.2 (The Verlet Method is 2nd Order) In this example, assume a single degree of freedom system, i.e. $q, p \in \mathbb{R}$, and take $M = 1$. The Verlet method can be written in the form of a map, as in (2.8), or, in slightly more detail, as

$$Q = q + hp + \frac{h^2}{2} F(q), \tag{2.13}$$

$$P = p + \frac{h}{2}\left[F(q) + F(q + hp + \frac{h^2}{2}F(q))\right]. \tag{2.14}$$

The first equation is already a polynomial, i.e. it is in the form of a series expansion in powers of h where the coefficients are functions of the starting point (q, p). The second equation may be written as a series expansion in powers of h as well:

$$P = p + \frac{h}{2}F(q) + \frac{h}{2}\left[F(q) + hF'(q)(p + \frac{h}{2}F(q)) + \frac{h^2}{2}F''(q)(p + \frac{h}{2}F(q))^2 + \ldots\right].$$

Note that the neglected terms will involve 4th (and higher) powers of h.